Update on form inference model

Contents

Update on form inference model 1
The context of the project 1
The problem of the form structure inference 1
The strategy 2
Current prototype 3
Represent analysis as a term 3
Example of Rewriting rules 3
Musical examples 4
Model output 4
Identifying presentation 4
example 2-11 4
example 2-12 5
Identifying compound themes 6
example 6-1 6
example 6-2 7
example 6-5 9
Others 11
supp 6-1 11
example 5-11 11
examples from annotation_pilot 11

The context of the project

We break the overarching form inference (model : piece at the note level → FormTree) into 3 parts

- harmonic reduction model : piece at the note level → harmony and cadence label
  - example output: `[I, V(HC), IV, V, I(PAC)]`
- motivic repetition model : piece at the note level → material markers
  - example output: `["a","b","c","a","b","c","d","e","a","b","c"]`
- **form structure model : (material marker, harmony, cadence) → FormTree**
  - example input:

    ```
    {a ⟦I-V None⟧} + {b ⟦I-III None⟧} + {a ⟦VI-II None⟧} + {b ⟦VI-III None⟧}
    ```

    we want to say that this is the same idea expressed in different harmonic context
  - example output:

    ```
    1 Presentation <b.i. := "a" + "b" >
    2       (idea usedOn ⟦I-V None⟧+⟦I-III None⟧)
    3       (idea usedOn ⟦VI-II None⟧+⟦VI-III None⟧)
    ```

The problem of the form structure inference

We want to describe the input in terms of formal units and variable binding in a compact way

- Formal function imposes constraints on what kind of analysis is possible
  - The term `(Presentation ideaBinding unit1 unit2)` entails that
    - unit1 begins on I and

  - the idea of unit1 and unit2 are the same
- Description length impose a ranking on alternative analysis (so the model should prefer the third instead of the first).
  - in general we can costs on the "operations" e.g. make "catch-all" merge rule such as `Concatenate` costly so the model will prefer to minimize their usage
  - the trivial description

```
1     = {a ⟦I-V None⟧} + {b ⟦I-III None⟧} + {a ⟦VI-II None⟧} + {b ⟦VI-III
  None⟧}
```

  - a short description

```
1     = (idea usedOn ⟦I-V None⟧+⟦I-III None⟧) + (idea usedOn ⟦VI-II None⟧+⟦VI-
  III None⟧)
2         where idea := "a" + "b"
```

  - a short description using formal functions

```
1     = Presentation <b.i. := "a" + "b" >
2         (b.i. usedOn ⟦I-V None⟧+⟦I-III None⟧)
3         (b.i. usedOn ⟦VI-II None⟧+⟦VI-III None⟧)
```

The strategy

- we represent form analysis as a term (of a recursive type).
- [mostly music-theoretical] we define term rewriting rules encoding music theoretical knowledge on form.
- we search for the smallest equivalent term for the input (reachable by the collection of rewrite rules). [mostly mathematical & programming]

Current prototype

Represent analysis as a term

Goals:

- we want to be able to recover the original surface (lossless encoding)
- we want to factor out formal repeats (compressible)
- we want the AST to be recursive, sentence and period should take FormalUnit as arguments (recursive)

The main AST

```hs
data FormalUnit
  = Leaf Idea PhraseParam
  | RefFormalUnit String
  | UseIdeaWith Idea PhraseParam
  | Sentence IdeaBinding FormalUnit FormalUnit
  | Period FormalUnitBinding FormalUnit FormalUnit
  | Presentation IdeaBinding FormalUnit FormalUnit
  | Continuation FormalUnit FormalUnit
  | Concatenate FormalUnit FormalUnit
  | ...

```

Auxillary definitions

```
data CadenceStrength = Weak | Strong | None
data Harmony = I | II | III | IV | V | VI | VII
data Idea
  = Label String
  | Concat Idea Idea
  | RefIdea String

data PhraseParam
  = Param [Harmony] CadenceStrength
  | ConcatParam PhraseParam PhraseParam

data IdeaBinding = BindIdea String Idea
data FormalUnitBinding = BindFormalUnit String FormalUnit
```

Note that FormalUnit is now a recursive AST

Example of Rewriting rules

```
premises
      match = Concatenate x y
      IdeaOf x = IdeaOf y
      StartingHarmonyOf x = I

conclusions
      match ~>
          (Presentation
              (BindIdea "b.i." (IdeaOf x))
              (UseIdeaWith (RefIdea "b.i.") (GetParam x))
              (UseIdeaWith (RefIdea "b.i.") (GetParam y))
          )
```

Musical examples

The musical examples, including scores and audios, links to the official Analyzing Classical Form website.

Among the https://acf.research.mcgill.ca/supp_list.phpcomplete list of example we look at how the form inference model perform on only a few of them.

Model output

Identifying presentation

example 2-11

🖼️🔊 **Click to Play**

We distinguish between bindings for idea and bindings for formal unit binding. For the presentation, it is the idea that is repeats, while for the period the repeating thing is the formal unit (ABAC)

```
Input surface:
size = 11
  {a ⟦I None⟧}
  {a ⟦IV None⟧}



Output analysis:
  size=11 compressionRate=1.0
  Seq
    idea = "a"
    parameters = ⟦I None⟧
                 ⟦IV None⟧


  size=11 compressionRate=1.0
  Concatenate
    {"a" ⟦I None⟧}
    {"a" ⟦IV None⟧}
```

This example illustrates the basics of how variable binding leads to smaller size. By default, we assign a high cost of raw repetition markers from the input (as they are placeholders for the surface motivic ideas). On the other hand, calling a binding variable just cost 1.

example 2-12

🖼️🔊 Click to Play

This time we make the input a little bit harder in that the basic idea is not an atomic marker, but the concatenation of two. And the model need to identify the repetition of “a”+“b” as the basic idea, and correctly build up the phrase parameter for the concatenated phrases.

```
Input surface:
size = 27
  {a ⟦I-V None⟧}
  {b ⟦I-III None⟧}
  {a ⟦VI-II None⟧}
  {b ⟦VI-III None⟧}



Output analysis:
  size=25 compressionRate=1.08
  Seq
    idea = "a"+"b"
    parameters = ConcatParam
                   ⟦I-V None⟧
                   ⟦I-III None⟧
                 ConcatParam
                   ⟦VI-II None⟧
                   ⟦VI-III None⟧


  size=27 compressionRate=1.0
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-III None⟧}
      Concatenate
        {"a" ⟦VI-II None⟧}
        {"b" ⟦VI-III None⟧}
```

Identifying compound themes

example 6-1

🖼️🔊 **Click to Play**

```
Input surface:
size = 60
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦IV-V-I None⟧}
  {d ⟦VI-VII-I-V Weak⟧}
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦V None⟧}
  {e ⟦I Strong⟧}



Output analysis:
  size=59 compressionRate=1.0169491525423728
  Period
    antecedentHead = "a"+"b"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦IV-V-I None⟧}
                     {"d" ⟦VI-VII-I-V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦V None⟧}
                     {"e" ⟦I Strong⟧}



  size=61 compressionRate=0.9836065573770492
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-V-I-V Weak⟧}
      Period
        antecedentHead = "c"
        antecedent = Concatenate
                       UseIdeaWith   $0 ⟦IV-V-I None⟧
                       Concatenate
                         {"d" ⟦VI-VII-I-V Weak⟧}
                         Concatenate
                           {"a" ⟦I-V None⟧}
                           {"b" ⟦I-V-I-V Weak⟧}
        consequent = Concatenate
                       UseIdeaWith   $0 ⟦V None⟧
                       {"e" ⟦I Strong⟧}
```

example 6-2

🖼️🔊 **Click to Play**

```
Input surface:
size = 64
  {a ⟦I None⟧}
  {b ⟦I None⟧}
  {c ⟦I None⟧}
  {a ⟦V None⟧}
  {b ⟦V None⟧}
  {c ⟦V None⟧}
  {b ⟦I-VI None⟧}
  {b ⟦IV-II None⟧}
  {b ⟦V None⟧}
  {d ⟦I-II-V-I Strong⟧}



Output analysis:
  size=61 compressionRate=1.0491803278688525
  Concatenate
    Seq
      idea = "a"+"b"+"c"
      parameters = ConcatParam
                     ⟦I None⟧
                     ConcatParam
                       ⟦I None⟧
                       ⟦I None⟧
                   ConcatParam
                     ⟦V None⟧
                     ConcatParam
                       ⟦V None⟧
                       ⟦V None⟧
    Sentence
      basicIdea = "b"
      presentation = Seq
                       idea = $0
                       parameters = ⟦I-VI None⟧
                                    ⟦IV-II None⟧
      continuation = Concatenate
                       {$0 ⟦V None⟧}
                       {"d" ⟦I-II-V-I Strong⟧}



  size=60 compressionRate=1.0666666666666667
  Concatenate
    Concatenate
      Seq
        idea = "a"+"b"+"c"
        parameters = ConcatParam
                       ⟦I None⟧
                       ConcatParam
```

```
                          ⟦I None⟧
                          ⟦I None⟧
                      ConcatParam
                        ⟦V None⟧
                        ConcatParam
                          ⟦V None⟧
                          ⟦V None⟧
      {"b" ⟦I-VI None⟧}
    Concatenate
      Seq
        idea = "b"
        parameters = ⟦IV-II None⟧
                     ⟦V None⟧
      {"d" ⟦I-II-V-I Strong⟧}
```

example 6-5

🖼️🔊 **Click to Play**

```
Input surface:
size = 92
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {d ⟦V Weak⟧}
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {e ⟦V-I Strong⟧}



Output analysis:
  size=80 compressionRate=1.15
  Period
    antecedentHead = Twice "a"+Twice "b"+Twice "c"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧
  ConcatParam   ⟦V-I None⟧ ⟦IV-I None⟧ ConcatParam   ⟦IV-I None⟧ ConcatParam   ⟦II
  None⟧ ⟦I None⟧
                   {"d" ⟦V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧
  ConcatParam   ⟦V-I None⟧ ⟦IV-I None⟧ ConcatParam   ⟦IV-I None⟧ ConcatParam   ⟦II
  None⟧ ⟦I None⟧
                   {"e" ⟦V-I Strong⟧}



  size=85 compressionRate=1.0823529411764705
  Period
    antecedentHead = Twice "a"+Twice "b"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧ ⟦V-I
  None⟧ ConcatParam   ⟦IV-I None⟧ ⟦IV-I None⟧
                   Concatenate
                     Seq
                       idea = "c"
                       parameters = ⟦II None⟧
                                    ⟦I None⟧
                     {"d" ⟦V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧ ⟦V-I
  None⟧ ConcatParam   ⟦IV-I None⟧ ⟦IV-I None⟧
```

```
                      Concatenate
                        Seq
                          idea = "c"
                          parameters = ⟦II None⟧
                                       ⟦I None⟧
                        {"e" ⟦V-I Strong⟧}
```

Others

supp 6-1

🖼️🔊 Click to Play

example 5-11

🖼️🔊 Click to Play

examples from annotation_pilot

```
Input surface:
size = 335
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
```

```
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}



Output analysis:
  size=146 compressionRate=2.2945205479452055
  SimpleBinary
    aPart = Period
              antecedentHead = "A"+"B"
              antecedent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             Concatenate
                               {"C" ⟦I None⟧}
                               {"D" ⟦I Weak⟧}
              consequent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             HybridAntCont
                               antecedent = Concatenate
                                              {"E" ⟦I None⟧}
                                              {"D" ⟦I Weak⟧}
                               continuation = Concatenate
                                                TwiceFormalUnit
                                                  {"A'" ⟦I None⟧}
                                                Concatenate
                                                  {"F" ⟦I None⟧}
                                                  {"G" ⟦I Strong⟧}
    bPart = Concatenate
              {"H" ⟦I None⟧}
              Concatenate
                {"I" ⟦I None⟧}
                Concatenate
                  {"J" ⟦I None⟧}
                  Concatenate
                    Concatenate
                      {"B" ⟦I Weak⟧}
                      Concatenate
                        TwiceFormalUnit
                          {"K" ⟦I None⟧}
                        Concatenate
                          {"L" ⟦I None⟧}
                          {"M" ⟦I Weak⟧}
                    Concatenate
```

```
                      TwiceFormalUnit
                        {"A'" ⟦I None⟧}
                      Concatenate
                        Concatenate
                          {"N" ⟦I None⟧}
                          {"G'" ⟦I Strong⟧}
                        Concatenate
                          TwiceFormalUnit
                            {"H'" ⟦I None⟧}
                          Concatenate
                            {"F" ⟦I None⟧}
                            {"G" ⟦I Strong⟧}


size=148 compressionRate=2.2635135135135136
Concatenate
  TwiceFormalUnit
    Period
      antecedentHead = "A"+"B"
      antecedent = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                     Concatenate
                       {"C" ⟦I None⟧}
                       {"D" ⟦I Weak⟧}
      consequent = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                     HybridAntCont
                       antecedent = Concatenate
                                      {"E" ⟦I None⟧}
                                      {"D" ⟦I Weak⟧}
                       continuation = Concatenate
                                        TwiceFormalUnit
                                          {"A'" ⟦I None⟧}
                                        Concatenate
                                          {"F" ⟦I None⟧}
                                          {"G" ⟦I Strong⟧}
  TwiceFormalUnit
    Concatenate
      {"H" ⟦I None⟧}
      Concatenate
        {"I" ⟦I None⟧}
        Concatenate
          {"J" ⟦I None⟧}
          Concatenate
            Concatenate
              {"B" ⟦I Weak⟧}
              Concatenate
                TwiceFormalUnit
                  {"K" ⟦I None⟧}
                Concatenate
                  {"L" ⟦I None⟧}
                  {"M" ⟦I Weak⟧}
            Concatenate
              TwiceFormalUnit
```

```
                  {"A'" ⟦I None⟧}
                Concatenate
                  Concatenate
                    {"N" ⟦I None⟧}
                    {"G'" ⟦I Strong⟧}
                  Concatenate
                    TwiceFormalUnit
                      {"H'" ⟦I None⟧}
                    Concatenate
                      {"F" ⟦I None⟧}
                      {"G" ⟦I Strong⟧}
```

```
Input surface:
size = 623
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
```

```
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}


```

```
Output analysis:
  size=154 compressionRate=4.045454545454546
  Ternary
    exposition = SimpleBinary
                   aPart = HybridAntCont
                             antecedent = Sentence
                                            basicIdea = "a"
                                            presentation = Seq
                                                             idea = $0
                                                             parameters = ⟦I
    None⟧
                                                                          ⟦V
    None⟧
                                            continuation = Concatenate
                                                             {$0 ⟦I None⟧}
                                                             {"b" ⟦V Weak⟧}
                             continuation = Concatenate
                                              TwiceFormalUnit
                                                {"c" ⟦I None⟧}
                                              Concatenate
                                                {"d" ⟦I None⟧}
                                                {"e" ⟦I Strong⟧}
                   bPart = HybridAntCont
                             antecedent = Concatenate
                                            Seq
                                              idea = "a"
                                              parameters = ⟦I None⟧
                                                           ⟦V None⟧
                                            Concatenate
                                              {"f" ⟦V None⟧}
                                              {"b" ⟦V Weak⟧}
                             continuation = Concatenate
                                              TwiceFormalUnit
                                                {"c" ⟦I None⟧}
                                              Concatenate
                                                {"d" ⟦V None⟧}
                                                {"e" ⟦I Strong⟧}
    contrastingMiddile = RoundedBinary
                           exposition = Concatenate
                                          Seq
                                            idea = "g"+"h"
                                            parameters = ConcatParam
                                                           ⟦I None⟧
                                                           ⟦I None⟧
                                                         ConcatParam
                                                           ⟦V None⟧
                                                           ⟦V None⟧
                                          Concatenate
                                            Concatenate
                                              {"g" ⟦I None⟧}
                                              {"h" ⟦IV None⟧}
                                            Concatenate
                                              {"i" ⟦V None⟧}
                                              {"j" ⟦I Strong⟧}
                           contrastingMiddile = Seq
```

```
                                                  idea = "g"+"k"
                                                  parameters = ConcatParam
                                                                 ⟦I None⟧
                                                                 ⟦IV None⟧
                                                               ConcatParam
                                                                 ⟦II None⟧
                                                                 ⟦V None⟧



size=201 compressionRate=3.099502487562189
Concatenate
  TwiceFormalUnit
    HybridAntCont
      antecedent = Sentence
                     basicIdea = "a"
                     presentation = Seq
                                      idea = $0
                                      parameters = ⟦I None⟧
                                                   ⟦V None⟧
                     continuation = Concatenate
                                      {$0 ⟦I None⟧}
                                      {"b" ⟦V Weak⟧}
      continuation = Concatenate
                       TwiceFormalUnit
                         {"c" ⟦I None⟧}
                       Concatenate
                         {"d" ⟦I None⟧}
                         {"e" ⟦I Strong⟧}
  Ternary
    exposition = TwiceFormalUnit
                   HybridAntCont
                     antecedent = Concatenate
                                    Seq
                                      idea = "a"
                                      parameters = ⟦I None⟧
                                                   ⟦V None⟧
                                    Concatenate
                                      {"f" ⟦V None⟧}
                                      {"b" ⟦V Weak⟧}
                     continuation = Concatenate
                                      TwiceFormalUnit
                                        {"c" ⟦I None⟧}
                                      Concatenate
                                        {"d" ⟦V None⟧}
                                        {"e" ⟦I Strong⟧}
    contrastingMiddile = Concatenate
                           RoundedBinary
                             exposition = Concatenate
                                            Seq
                                              idea = "g"+"h"
                                              parameters = ConcatParam
                                                             ⟦I None⟧
                                                             ⟦I None⟧
                                                           ConcatParam
```

```
                                                                  ⟦V None⟧
                                                                  ⟦V None⟧
                                                Concatenate
                                                  Concatenate
                                                    {"g" ⟦I None⟧}
                                                    {"h" ⟦IV None⟧}
                                                  Concatenate
                                                    {"i" ⟦V None⟧}
                                                    {"j" ⟦I Strong⟧}
                                    contrastingMiddile = Seq
                                                         idea = "g"+"k"
                                                         parameters = ConcatParam
                                                                        ⟦I None⟧
                                                                        ⟦IV None⟧
                                                                      ConcatParam
                                                                        ⟦II None⟧
                                                                        ⟦V None⟧
                                  TwiceFormalUnit
                                    HybridAntCont
                                      antecedent = Sentence
                                                   basicIdea = "a"
                                                   presentation = Seq
                                                                  idea = $0
                                                                  parameters = ⟦I
    None⟧
                                                                                ⟦V
    None⟧
                                                   continuation = Concatenate
                                                                  {$0 ⟦I None⟧}
                                                                  {"b" ⟦V Weak⟧}
                                      continuation = Concatenate
                                                     TwiceFormalUnit
                                                       {"c" ⟦I None⟧}
                                                     Concatenate
                                                       {"d" ⟦I None⟧}
                                                       {"e" ⟦I Strong⟧}
```